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Quaternion-based nonlinear attitude control of quadrotor formations carrying a slung load

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ABSTRACT

Control of a pair of quadrotors carrying a slung load is addressed. A hierarchical control algorithm is implemented on the leader quadrotor. The control system features a linear quadratic tracking controller on its outer loop for velocity command tracking and generates the thrust vector commands for the inner loop. The inner loop uses the *to-go* quaternion calculated from the thrust vector commands to control the attitude. The second quadrotor, designated as the follower quadrotor, has another hierarchical control structure on it. This control system features a Lyapunov function based nonlinear controller for formation guidance and thrust vector command generation in the outer loop. In the inner loop, the same nonlinear attitude controller is used. The dynamics of the two quadrotor slung load system is modelled and two simulation codes are developed utilizing first rigid, and then flexible load carrying rods. Simulation results are presented to show the efficacy of the control algorithm to keep the two quadrotors flying in a fixed geometrical formation in spite of the unmodelled disturbance dynamics of both the rigid bars, flexible bars and the load.

1. Introduction

A quadrotor is a rotary-wing UAV with hovering and vertical take-off and landing (VTOL) capabilities. It has two pairs of fixed-pitch propellers with one pair rotating in a clockwise direction and the other pair in a counterclockwise direction. These propellers generate a thrust vector that counteracts the effects of gravity force on the quadrotor. The thrust vector direction of the quadrotor is attitude dependent and can be controlled by creating differential thrusts in the pair of clockwise and counterclockwise rotating propellers respectively. Hence, the dynamics of a quadrotor is much simpler than other VTOL aircrafts like the helicopter [1].

Multirotor platforms such as the quadrotors find their use in various missions such as search and rescue, surveillance and environment monitoring. Multirotor platforms are also used for cargo transportation, deployment of sensors and load drop missions [2,3] which is the focus of the work presented in this paper. There are mainly two approaches in literature to tackle the load transportation problem. The first is attaching a robotic gripper to the flying vehicle and the second is the slung (Suspended) load approach. The advantage of the slung load over the gripper is reduced mass and inertia of the system, thereby providing a better transient attitude response [4,5].

Recently, it was requested to develop autonomous quadrotors, to be used for search and rescue operations, delivering urgently needed packages to the people stranded in nature. The quadrotors takeoff weight shall less than 4 kg, dictated by the regulations imposed in the country for “Level 0” certified pilots. In addition, a 1 kg payload was to be carried for at least 45 minutes. These requirements were successfully realized by a careful selection of the propellers, brushless dc-motors, motor drivers, and batteries [6]. However, it soon became clear that, depending on the search and rescue operation, it may be necessary to carry larger payloads. One solution to this problem is to rapidly reprogram the same identical quadrotors to fly together to carry a larger payload in a cooperative fashion. Since the search and rescue team will not have any knowledge about the inertia properties, or the dynamics of the load carrying system, the controller shall be capable of handling such wide range of payloads. There will be only two separate controller algorithms to be loaded before take-off: one of the leader quadrotor, and another one for the followers. In this way, the search and rescue team may easily configure the system, using standard parts, at the base station, which is the motivation of the work presented in this manuscript.

The slung load system significantly alters the flying characteristics of the UAV, presenting a challenge in controlling the UAV. The

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Nomenclature

J_i, J_{c_i}, J_L	Inertia Matrix of quadrotor i, cable i and Load
$\boldsymbol{\omega}_i$	Quadrotor i angular velocity, written in quadrotor i fixed frame
$\boldsymbol{\omega}_{c_i}$	Cable i angular velocity, written in cable i fixed frame
$\boldsymbol{\omega}_L$	Cable i angular velocity, written in Load fixed frame
C_i^N	the direction cosine matrix from the inertial frame to quadrotor i frame
C_L^N	the direction cosine matrix from the inertial frame to the load frame.
$C_{c_i}^N$	the direction cosine matrix from the inertial frame to cable i frame.
$\boldsymbol{F}_{si}$	the reaction force at the spherical joints connecting the quadrotors, cables and Load
$\boldsymbol{T}_{si}$	friction and damping torques at the joints.
T_x, T_y, T_z	Quadrotor Control Input Torques
d	Quadrotor Level Arm
Ω_i	Angular velocity of Quadrotor Propeller i
k_m	Constant that relates thrust generated by the propeller to its Torque
k_n	Constant that relates thrust generated by the propeller to its angular velocity
$\omega_1^\times$	Matrix to carry out cross product: $\omega_1^\times r_1 = \omega_1 \times r_1$
$\omega_1^{\times\times}$	Matrix to carry out cross double product: $\omega_1^{\times\times} r_1 = \omega_1 \times (\omega_1 \times r_1)$
I_x, I_y, I_z	Moment of Inertia along the x, y and z axis respectively
$\mathbf{t}$	Vectorial part of the *to-go* quaternion
t_4	Scalar part of the *to-go* quaternion

additional nonactuated degrees of freedom related to the swinging and rocking of the slung load also significantly complicates the equations of motion of the system presenting a challenge in obtaining a mathematical model for simulation purposes. The slung load approach to load transportation introduces non-actuated degrees of freedom into the system model due to the swinging and rocking of the load. There have been a number of approaches in literature to tackle this problem. The control strategies for the slung load problem falls into either generating swing free maneuvers or the stabilization of the swing dynamics. A dynamic programming approach for a trajectory generation of a swing free maneuver of a quadrotor slung long system is employed in [2]. In [7,8], various input shaping controls to reduce the load swing angle are investigated. An adaptive controller to cope with the changes in the centre of gravity of the UAV as the load swings is designed [2]. In [9], the position of the payload relative to the quadrotor is estimated through a downward-facing camera. The trajectory of the load is also preplanned. A controller that utilises the payload swings is then implemented. In [10], a state observed to estimate the disturbance introduced into the system as a result of the payload motion is designed. A disturbance rejection controller is then implemented.

An alternative approach to limit the swinging dynamics of the slung load is to use multiple UAVs to cooperatively carry a single load. The additional bars/cables used to connect the UAVs to the load inherently introduce additional constraints to the motion of the payload [11]. Multiple UAVs also gives way to carry a larger payload. In [12], two PID controllers were designed. One PID controller to control the motion of the quadrotor and the other to control the swing of the load. A desired trajectory for the quadrotors working cooperatively are then obtained from the desired trajectory of the load. Many of the available control techniques involve using various motion tracking systems, real-time kinematic devices or on-board camera for the UAV and load position estimation. This leads to the need of additional sensors to be attached to the load requiring a reduction in the available payload weight. In this paper, the load dynamics is treated as an unmodelled dynamics as well as a disturbance to our control algorithm removing the need to accurately determine the motion of the load. To further limit the load deflection, additional constraints are introduced into the system through a formation guidance algorithm implemented on the two quadrotors carrying the load [13,14]. A nonlinear cascaded Lyapunov function based algorithm is implemented to control the motion of the quadrotor in spite of the unmodelled dynamics of the slung load. The control of the quadrotors and the formation guidance is done in the outer loop, while attitude control is implemented in the inner loop. The control scheme ensures that the quadrotors and payload converge to the same velocity without explicit communication among them.

There are a number of published works in literature on the control of quadrotors. Some of these control approaches include sliding mode control [15,16], backstepping technique [17], PID controllers as well as linear quadratic control techniques [18], adaptive control [19] and fuzzy logic control [20]. The utilization of quaternions for quadrotor attitude control is becoming quite popular in literature. There are a number of advantages of using quaternions. For attitude propagation, singularities are avoided and trigonometric function evaluations are not required. Using quaternion parameterization is commonly used in the attitude control of Earth observation satellites when large slew angle maneuvers are needed [21]. There have been a number of approaches proposed in literature for the attitude control of quadrotors using quaternions [22–25]. In [24], Tayebi uses a Lyapunov function that contained body angular velocities and quaternions. The attitude command signals in terms of roll, pitch and yaw commands were calculated from a position control algorithm and then converted to a quaternion command for the attitude control system. In [23], Reyes et al., also use quaternions for attitude parameterization. A linear gain scheduled controller is then employed. In [22], Euler angles are used for attitude control and a sliding mode estimator and controller is used for the translational motion in a body fixed frame coordinate. In [25], the necessary thrust vector for the position control over a prescribed trajectory is first calculated, and then converted to an attitude trajectory by solving some nonlinear algebraic equations. Attitude control is then carried out using this trajectory via a linear controller. To use this linear controller, a new variable, which is the natural logarithm of the quaternions was used. The modelling and control of a two quadrotor slung load system as well as the guidance and control of aircraft flying in formation has been addressed previously [26,27].

The main contributions of this work may be summarised as follows: First, the controllers do not measure the motion of the slung load, or aware of the inertial characteristics of the payload. Second, only outer loop controller of the leader and follower(s) are different. The leader is commanded by the pilot, and the followers track the leader, by using its position and velocity information, through a Lyapunov function based formation controller. Finally, different from other quadrotor attitude controllers proposed in the literature, in the inner loop, a novel quaternion based nonlinear attitude controller is used.

In the next sections, two quadrotors with a slung load modelling is given first. Two models are presented. In the first model, rigid load carrying bars are considered. The second model is obtained from ADAMS software, where the bars are flexible. The

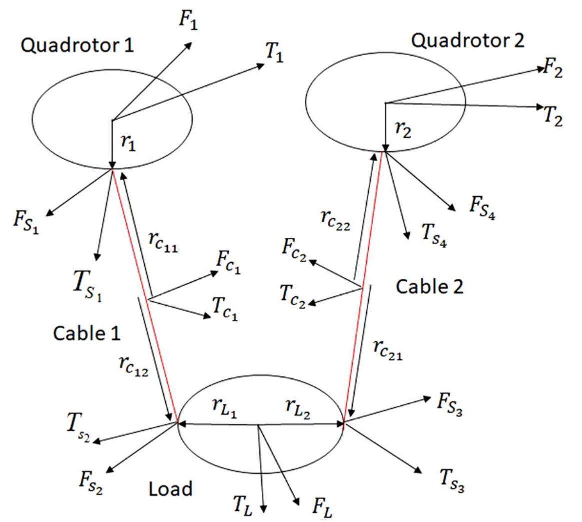

Fig. 1. Two quadrotor slung load system.

control algorithms are presented next. Then, simulation results are given and discussed. Finally conclusions are given.

2. Dynamical equations for the two-quadrotor slung load system

This paper presents a straightforward, easily extendable formulation of the mathematical model of a two quadrotor slung load system with an efficient computer implementation. The formulation is an extension to the approach presented by Stoneking [28] for modelling multi-body spacecrafts. The building block equations are derived by applying Newton's and Euler's equations of motion to an "element" consisting of five bodies and four joints. In this case, the five bodies are the two quadrotors, the two rigid bars connecting the quadrotors to the load and the load itself; the four joints are spherical joints through which the bars connect the quadrotors to the load. The mathematical model presents a realistic full nonlinear 3-D model of the entire system, as the quadrotors, rigid bars and load all have masses and inertias. The suspension point in the mathematical model is also not at the centre of mass of the quadrotors. The bars are also connected to the load at two separate points through spherical joints. Fig. 1.

In modelling the quadrotor slung load system, it is assumed that the bars connecting the load to the quadrotors are stiff. Spherical joints connect the bars to the quadrotor and load and thus the joint torques are assumed to be zero. The rotational equations for the two quadrotors in their own respective quadrotor fixed reference frame can be written as [27]:

$$
\begin{aligned}
J_1\dot{\boldsymbol{\omega}}_1 &= \boldsymbol{T}_1 + \boldsymbol{T}_{S_1} - \omega_1^\times J_1\boldsymbol{\omega}_1 + r_1^\times C_N^1 \boldsymbol{F}_{S_1}\\
J_2\dot{\boldsymbol{\omega}}_2 &= \boldsymbol{T}_2 - C_{c_2}^2 \boldsymbol{T}_{S_4} - r_2^\times C_N^2 \boldsymbol{F}_{S_4} - \omega_2^\times J_2\boldsymbol{\omega}_2
\end{aligned}
\tag{1}
$$

Similarly, the rotational equations of motion of the load and cables in their own respective load and cable fixed frame of reference can be written as:

$$
\begin{aligned}
J_{c_1}\dot{\boldsymbol{\omega}}_{c_1} =& \boldsymbol{T}_{c_1} - C_1^{c_1}\boldsymbol{T}_{S_1} - r_{c_{11}}^\times C_N^{c_1}\boldsymbol{F}_{S_1}\\
& + \boldsymbol{T}_{s_2} + r_{c_{12}}^\times C_N^{c_1}\boldsymbol{F}_{S_2} - \omega_{c_1}^\times J_{c_1}\boldsymbol{\omega}_{c_1}\\
J_L\dot{\boldsymbol{\omega}}_L =& \boldsymbol{T}_L - C_{c_1}^L\boldsymbol{T}_{s_2} - r_{L_1}^\times C_N^L\boldsymbol{F}_{S_2} + \boldsymbol{T}_{s_3} + r_{L_2}^\times C_N^L\boldsymbol{F}_{S_3} - \omega_L^\times J_L\boldsymbol{\omega}_L\\
J_{c_2}\dot{\boldsymbol{\omega}}_{c_2} =& \boldsymbol{T}_{c_2} - C_L^{c_2}\boldsymbol{T}_{S_3} - r_{c_{21}}^\times C_N^{c_2}\boldsymbol{F}_{S_3} + \boldsymbol{T}_{s_4}\\
& + r_{c_{22}}^\times C_N^{c_2}\boldsymbol{F}_{S_4} - \omega_{c_2}^\times J_{c_2}\boldsymbol{\omega}_{c_2}
\end{aligned}
\tag{2}
$$

The translational equations written in the inertial frame of reference are:

$$
\begin{aligned}
m_1\dot{\boldsymbol{v}}_1 &= \boldsymbol{F}_1 + \boldsymbol{F}_{S_1}\\
m_{c_1}\dot{\boldsymbol{v}}_{c_1} &= \boldsymbol{F}_{c_1} + \boldsymbol{F}_{S_2} - \boldsymbol{F}_{S_1}\\
m_L\dot{\boldsymbol{v}}_L &= \boldsymbol{F}_L + \boldsymbol{F}_{S_3} - \boldsymbol{F}_{S_2}\\
m_{c_2}\dot{\boldsymbol{v}}_{c_2} &= \boldsymbol{F}_{c_2} + \boldsymbol{F}_{S_4} - \boldsymbol{F}_{S_3}\\
m_2\dot{\boldsymbol{v}}_2 &= \boldsymbol{F}_2 - \boldsymbol{F}_{S_4}
\end{aligned}
\tag{3}
$$

The constraint equations can then be obtained by equating the joint accelerations:

$$
\begin{aligned}
&\boldsymbol{v}_{S_1} = \boldsymbol{v}_1 + \omega_1^\times\boldsymbol{r}_1 = \boldsymbol{v}_{c_1} + \omega_{c_1}^\times\boldsymbol{r}_{c_{11}}\\
&\dot{\boldsymbol{v}}_1 + C_1^N\dot{\omega}_1^\times\boldsymbol{r}_1 + C_1^N\omega_1^{\times\times}\boldsymbol{r}_1 = \dot{\boldsymbol{v}}_{c_1} + C_{c_1}^N\dot{\omega}_{c_1}^\times\boldsymbol{r}_{c_{11}} + C_{c_1}^N\omega_{c_1}^{\times\times}\boldsymbol{r}_{c_{11}}\\
&\Rightarrow \dot{\boldsymbol{v}}_{c_1} - \dot{\boldsymbol{v}}_1 = C_1^N\dot{\omega}_1^\times\boldsymbol{r}_1 + C_1^N\omega_1^{\times\times}\boldsymbol{r}_1 - C_{c_1}^N\dot{\omega}_{c_1}^\times\boldsymbol{r}_{c_{11}} - C_{c_1}^N\omega_{c_1}^{\times\times}\boldsymbol{r}_{c_{11}}\\
&\boldsymbol{v}_{S_2} = \boldsymbol{v}_{c_1} + \omega_{c_1}^\times\boldsymbol{r}_{c_{12}} = \boldsymbol{v}_L + \omega_L^\times\boldsymbol{r}_{L_1}\\
&\dot{\boldsymbol{v}}_{c_1} + C_{c_1}^N\dot{\omega}_{c_1}^\times\boldsymbol{r}_{c_{12}} + C_{c_1}^N\omega_{c_1}^{\times\times}\boldsymbol{r}_{c_{12}} = \dot{\boldsymbol{v}}_L + C_L^N\dot{\omega}_L^\times\boldsymbol{r}_{L_1} + C_L^N\omega_L^{\times\times}\boldsymbol{r}_{L_1}\\
&\Rightarrow \dot{\boldsymbol{v}}_L - \dot{\boldsymbol{v}}_{c_1} = C_{c_1}^N\dot{\omega}_{c_1}^\times\boldsymbol{r}_{c_{12}} + C_{c_1}^N\omega_{c_1}^{\times\times}\boldsymbol{r}_{c_{12}} - C_L^N\dot{\omega}_L^\times\boldsymbol{r}_{L_1} - C_L^N\omega_L^{\times\times}\boldsymbol{r}_{L_1}\\
&\boldsymbol{v}_{S_3} = \boldsymbol{v}_L + \omega_L^\times\boldsymbol{r}_{L_2} = \boldsymbol{v}_{c_2} + \omega_{c_2}^\times\boldsymbol{r}_{c_{21}}\\
&\dot{\boldsymbol{v}}_L + C_L^N\dot{\omega}_L^\times\boldsymbol{r}_{L_2} + C_L^N\omega_L^{\times\times}\boldsymbol{r}_{L_2} = \dot{\boldsymbol{v}}_{c_2} + C_{c_2}^N\dot{\omega}_{c_2}^\times\boldsymbol{r}_{c_{21}} + C_{c_2}^N\omega_{c_2}^{\times\times}\boldsymbol{r}_{c_{21}}\\
&\Rightarrow \dot{\boldsymbol{v}}_{c_2} - \dot{\boldsymbol{v}}_L = C_L^N\dot{\omega}_L^\times\boldsymbol{r}_{L_2} + C_L^N\omega_L^{\times\times}\boldsymbol{r}_{L_2} - C_{c_2}^N\dot{\omega}_{c_2}^\times\boldsymbol{r}_{c_{21}} - C_{c_2}^N\omega_{c_2}^{\times\times}\boldsymbol{r}_{c_{21}}\\
&\boldsymbol{v}_{S_4} = \boldsymbol{v}_{c_2} + \omega_{c_2}^\times\boldsymbol{r}_{c_{22}} = \boldsymbol{v}_2 + \omega_2^\times r_2\\
&\dot{\boldsymbol{v}}_{c_2} + C_{c_2}^N\dot{\omega}_{c_2}^\times\boldsymbol{r}_{c_{22}} + C_{c_2}^N\omega_{c_2}^{\times\times}\boldsymbol{r}_{c_{22}} = \dot{\boldsymbol{v}}_2 + C_2^N\dot{\omega}_2^\times\boldsymbol{r}_2 + C_2^N\omega_2^{\times\times}\boldsymbol{r}_2\\
&\Rightarrow \dot{\boldsymbol{v}}_2 - \dot{\boldsymbol{v}}_{c_2} = C_{c_2}^N\dot{\omega}_{c_2}^\times\boldsymbol{r}_{c_{22}} + C_{c_2}^N\omega_{c_2}^{\times\times}\boldsymbol{r}_{c_{22}} - C_2^N\dot{\omega}_2^\times\boldsymbol{r}_2 - C_2^N\omega_2^{\times\times}\boldsymbol{r}_2
\end{aligned}
\tag{4}
$$

In vector-matrix form, all the above equations may be written as:

$$
\begin{bmatrix} A & 0 & R\\ 0 & M & U\\ R^T & U^T & 0 \end{bmatrix}
\begin{Bmatrix} \dot{\boldsymbol{x}}\\ \dot{\boldsymbol{y}}\\ \boldsymbol{f} \end{Bmatrix}
=
\begin{Bmatrix} \boldsymbol{\tau}\\ \boldsymbol{\upsilon}\\ \boldsymbol{\vartheta} \end{Bmatrix}
\tag{5}
$$

where

$$
A = \begin{bmatrix}
J_1 & 0 & 0 & 0 & 0 & 0\\
0 & J_{c_1} & 0 & 0 & 0 & 0\\
0 & 0 & J_L & 0 & 0 & 0\\
0 & 0 & 0 & J_{c_2} & 0 & 0\\
0 & 0 & 0 & 0 & J_2 & 0\\
0 & 0 & 0 & 0 & 0 & m_1 I
\end{bmatrix},
$$

$$
R = \begin{bmatrix}
-r_1^\times C_N^1 & 0 & 0 & 0\\
r_{c_{11}}^\times C_N^{c_1} & -r_{c_{12}}^\times C_N^{c_1} & 0 & 0\\
0 & r_{L_1}^\times C_N^L & -r_{L_2}^\times C_N^L & 0\\
0 & 0 & r_{c_{21}}^\times C_N^{c_2} & r_{c_{22}}^\times C_N^{c_2}\\
0 & 0 & 0 & r_2^\times C_N^2\\
-I & 0 & 0 & 0
\end{bmatrix}
\tag{6}
$$

$$
M = \begin{bmatrix}
m_{c_1} I & 0 & 0 & 0\\
0 & m_L I & 0 & 0\\
0 & 0 & m_{c_2} I & 0\\
0 & 0 & 0 & m_2 I
\end{bmatrix}, \quad
U = \begin{bmatrix}
I & -I & 0 & 0\\
0 & I & -I & 0\\
0 & 0 & I & -I\\
0 & 0 & 0 & I
\end{bmatrix}
\tag{7}
$$

$$
\boldsymbol{\tau} = \begin{Bmatrix} \boldsymbol{T}_1 - \omega_1^{\times} J_1 \boldsymbol{\omega}_1 + \boldsymbol{T}_{S_1} \\ \boldsymbol{T}_{c_1} - \omega_{c_1}^{\times} J_{c1} \boldsymbol{\omega}_{c_1} - C_1^{c_1} \boldsymbol{T}_{S_1} + \boldsymbol{T}_{s_2} \\ \boldsymbol{T}_L - \omega_L^{\times} J_L \boldsymbol{\omega}_L - C_{c_1}^{L} \boldsymbol{T}_{s_2} + \boldsymbol{T}_{s_3} \\ \boldsymbol{T}_{c_2} - \omega_{c_2}^{\times} J_{c2} \boldsymbol{\omega}_{c_2} - C_L^{c_2} \boldsymbol{T}_{S_3} + \boldsymbol{T}_{s_4} \\ \boldsymbol{T}_2 - \omega_2^{\times} J_2 \boldsymbol{\omega}_2 - C_{c_2}^{2} \boldsymbol{T}_{S_4} \\ F_1 \end{Bmatrix}, \tag{8}
$$

$$
\boldsymbol{\upsilon} = \begin{Bmatrix} \boldsymbol{F}_{c_1} \\ \boldsymbol{F}_L \\ \boldsymbol{F}_{c_2} \\ \boldsymbol{F}_2 \end{Bmatrix}, \quad \boldsymbol{\vartheta} = \begin{Bmatrix} C_1^N \omega_1^{\times\times} \boldsymbol{r}_1 - C_{c_1}^N \omega_{c_1}^{\times\times} \boldsymbol{r}_{c_{11}} \\ C_{c_1}^N \omega_{c_1}^{\times\times} \boldsymbol{r}_{c_{12}} - C_L^N \omega_L^{\times\times} \boldsymbol{r}_{L_1} \\ C_L^N \omega_L^{\times\times} \boldsymbol{r}_{L_2} - C_{c_2}^N \omega_{c_2}^{\times\times} \boldsymbol{r}_{c_{21}} \\ C_{c_2}^N \omega_{c_2}^{\times\times} \boldsymbol{r}_{c_{22}} - C_2^N \omega_2^{\times\times} \boldsymbol{r}_2 \end{Bmatrix}
$$

$$
\dot{\boldsymbol{x}} = \begin{Bmatrix} \dot{\boldsymbol{\omega}}_1 \\ \dot{\boldsymbol{\omega}}_{c_1} \\ \dot{\boldsymbol{\omega}}_L \\ \dot{\boldsymbol{\omega}}_{c_2} \\ \dot{\boldsymbol{\omega}}_2 \\ \dot{\boldsymbol{v}}_1 \end{Bmatrix}, \quad \dot{\boldsymbol{y}} = \begin{Bmatrix} \dot{\boldsymbol{v}}_{c_1} \\ \dot{\boldsymbol{v}}_L \\ \dot{\boldsymbol{v}}_{c_2} \\ \dot{\boldsymbol{v}}_2 \end{Bmatrix}, \quad \boldsymbol{f} = \begin{Bmatrix} \boldsymbol{F}_{S_1} \\ \boldsymbol{F}_{S_2} \\ \boldsymbol{F}_{S_3} \\ \boldsymbol{F}_{S_4} \end{Bmatrix} \tag{9}
$$

$\dot{x}$, $\dot{y}$ and f can be decoupled in Eq. (5) by introducing unknown coefficient matrices α and β and performing row operations:

$$
\left(A + \beta R^T\right)\dot{\boldsymbol{x}} + \left(\alpha M + \beta U^T\right)\dot{\boldsymbol{y}} + (R + \alpha U)\,\boldsymbol{f} = \boldsymbol{\tau} + \alpha\boldsymbol{\upsilon} + \beta\boldsymbol{\vartheta} \tag{10}
$$

choose α = $-RU^{-1}$, $\beta = RU^{-1}MU^{-T}$, then,

$$
\left(A + \beta R^T\right)\dot{\boldsymbol{x}} = \boldsymbol{\tau} + \alpha\boldsymbol{\upsilon} + \beta\boldsymbol{\vartheta} \tag{11}
$$

Once the above Eq. (11) is solved for $\dot{\boldsymbol{x}}$, $\dot{\boldsymbol{y}}$ and $\boldsymbol{f}$ can then be easily computed from Eq. (5).

2.1. Extension to a two quadrotor formation carrying a slung load using flexible bars

The quadrotor slung load system with flexible bars is modelled using MSC ADAMS Software. The system is a 5-body model consisting of two quadrotors, two aluminium flexible bars and a load. The 5-bodies are connected to each other via a total of 4 spherical joints. This model is then imported into the MATLAB/SIMULINK platform. The two quadrotors in the system receive force and torque as inputs, and the system outputs the translational and angular velocities of the 5 bodies in the system. Using these respective angular velocities, the respective transformation matrices to the inertial frame are propagated. From the transformation matrices for each body, the respective Euler angles are derived. Thus a full dynamical model for the slung load system is obtained. The developed control algorithms are then tested on this model. Fig. 2.

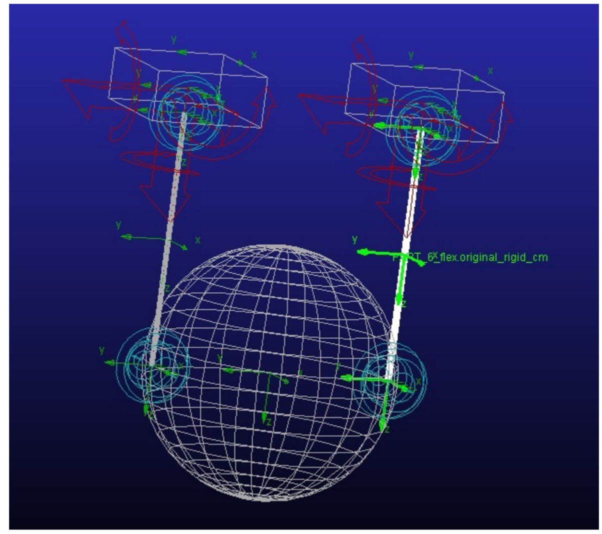

Fig. 2. Two quadrotors slung load system with flexible beam ADAMS.

3. Formulation of the control problem

On the designated leader quadrotor, a hierarchical two loop control system is implemented. The outer loop of this control system features a linear quadratic tracking (LQT) velocity controller that tracks reference velocity commands sent out by the operator and generates the control force command (thrust vector) that the leader quadrotor must track to achieve the desired velocity commands. This generated control force command is then sent to the inner loop of the control system. Through the pitching and rolling motion of the quadrotor, the inner loop orients the total force acting on the quadrotor to match the control force command coming from the outer loop. The inner loop controller is designed using a Lyapunov function based algorithm that uses the *to-go* quaternion. Fig. 3.

On the designated follower, another hierarchical two loop control system is implemented. The outer loop of this control system features a Lyapunov function based formation guidance controller. This outer loop controller receives the leader quadrotor velocities as well as the relative position between the quadrotors and generates a control force command (thrust vector) that the follower must track to achieve the desired relative position from the leader quadrotor. This generated control force is also sent into the inner loop of the control system. Fig. 4.

3.1. Leader quadrotor outer loop LQT velocity controller

Given the following linear completely controllable and observable system: [29]

$$
\begin{aligned} \dot{\mathbf{x}}(t) &= A\mathbf{x}(t) + B\mathbf{u}(t) + \mathbf{f}(t) \\ \mathbf{y}(t) &= C\mathbf{x}(t) \end{aligned} \tag{12}
$$

and a performance index:

$$
J = \frac{1}{2}\int_0^{\infty} \left\{\mathbf{e}^t(t)Q(t)\mathbf{e}(t) + \mathbf{u}^t(t)R(t)\mathbf{u}(t)\right\}]dt \tag{13}
$$

where the desired output $\mathbf{z}(t)$ and the error $\mathbf{e}(t) = \mathbf{z}(t) - \mathbf{y}(t)$ yield the following algebraic Ricatti and auxiliary equation to be solved:

$$
-PA - A^tP + PBR^{-1}B^tP - C^tQC = 0 \tag{14}
$$

Then the linear quadratic tracking (LQT) optimal control law is found as [29]:

$$
\mathbf{u}(t) = K\mathbf{x}(t) + K_Z\mathbf{z}(t) + K_f\mathbf{f}(t) \tag{15}
$$

where the corresponding controller gains are:

$$
\begin{aligned} K &= -R^{-1}B^tP \\ K_Z &= R^{-1}B^t[PE - A^t]^{-1}W \\ K_f &= -R^{-1}B^t[PE - A^t]^{-1}P \end{aligned} \tag{16}
$$

and

$$
E = BR^{-1}B^t \tag{17}
$$

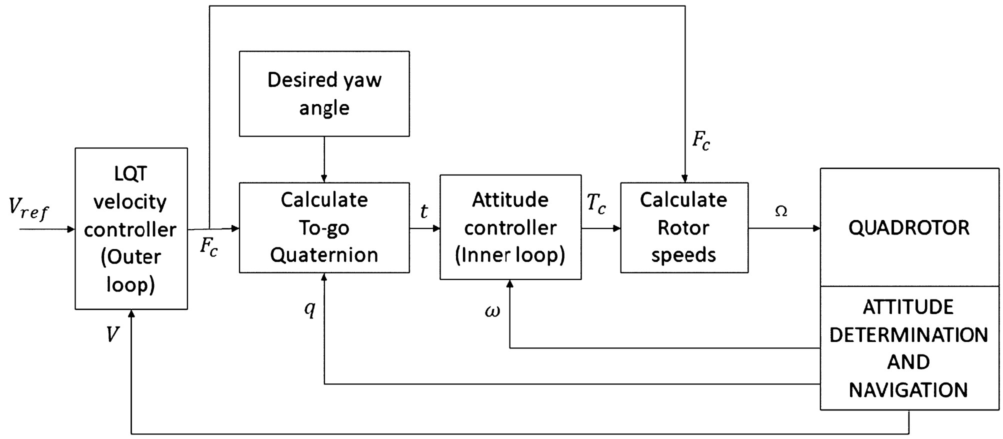

Fig. 3. Leader quadrotor control system schematic.

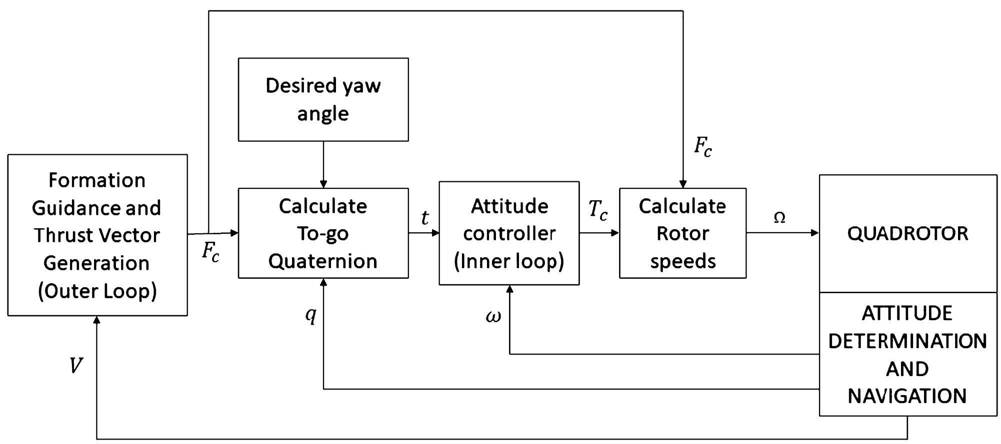

Fig. 4. Follower quadrotor control system schematic.

$$W = C^t Q \tag{18}$$

The LQT velocity controller should receive velocity commands and output the required thrust vector magnitude and direction to track the velocity commands. To do this, the translational equation of the quadrotor, written in the inertial frame are:

$$\begin{aligned}\dot{v}_x &= \frac{-k_x}{m} v_x + \frac{1}{m} F_x \\ \dot{v}_y &= \frac{-k_y}{m} v_y + \frac{1}{m} F_y \\ \dot{v}_z &= \frac{-k_z}{m} v_z + \frac{1}{m} F_z + g\end{aligned} \tag{19}$$

Eq. (19) can then be written in state-space form as:

$$\begin{Bmatrix} \dot{v}_x \\ \dot{v}_y \\ \dot{v}_z \end{Bmatrix} = \begin{bmatrix} \frac{-k_x}{m} & 0 & 0 \\ 0 & \frac{-k_y}{m} & 0 \\ 0 & 0 & \frac{-k_z}{m} \end{bmatrix} \begin{Bmatrix} v_x \\ v_y \\ v_z \end{Bmatrix} + \begin{bmatrix} \frac{1}{m} & 0 & 0 \\ 0 & \frac{1}{m} & 0 \\ 0 & 0 & \frac{1}{m} \end{bmatrix} \begin{Bmatrix} F_x \\ F_y \\ F_z \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ g \end{Bmatrix} \tag{20}$$

Using Eq. (20), the controller gains in Eq. (16) can be obtained. Note that in the above equations, disturbances associated with the bar-slung load system are neglected.

3.2. Follower quadrotor outer loop Lyapunov function based formation guidance controller

The Lyapunov stability theorem states that: Let $x = 0$ be an equilibrium point of a nonlinear system $\dot{x} = f(x)$. Let $L : D \to R$ be a continuously differentiable function in the neighbourhood D of $x = 0$, such that $L(0) = 0$ and $L(x) > 0$ in $D - \{0\}$. If $\dot{L}(x) \le 0$, then $x = 0$ is stable. Moreover, if $\dot{V}(x) < 0$ in $D - \{0\}$, then $x = 0$ is asymptotically stable [30].

The Lyapunov stability theorem approach may be used to design stabilizing controllers for nonlinear systems [30]. Our formation guidance controller is thus designed using the Lyapunov stability theorem. We used this approach to design a controller that guides our follower quadrotor to maintain a desired relative position from the leader quadrotor. Consider the following Lyapunov function:

$$L = \frac{1}{2} \Delta \mathbf{v}^T m_F \Delta \mathbf{v} + \frac{1}{2} \Delta \mathbf{r}^T R \Delta \mathbf{r} \tag{21}$$

where

$$\Delta \mathbf{r} = \begin{Bmatrix} x_L + a - x_F \\ y_L + b - y_F \\ z_L + c - z_F \end{Bmatrix}, \Delta \mathbf{v} = \begin{Bmatrix} v_{x_L} - v_{x_F} \\ v_{y_L} - v_{y_F} \\ v_{z_L} - v_{z_F} \end{Bmatrix} \tag{22}$$

(x_L, y_L, z_L) and (x_F, y_F, z_F) are the positions of the leader and the follower quadrotors with respect to the inertial frame. (a, b, c) is the desired relative position of the follower with respect to the leader quadrotor. $(v_{x_L}, v_{y_L}, v_{z_L})$ and $(v_{x_F}, v_{y_F}, v_{z_F})$ are the velocities of the leader and follower quadrotors with respect to the inertial frame. R is a positive definite matrix and m_F is the mass of the follower quadrotor. Assuming a fixed formation geometry, i.e., $(\dot{a}, \dot{b}, \dot{c}) = 0$, the derivative of the Lyapunov function gives:

$$\dot{L} = \Delta\mathbf{v}^T m_F \Delta\dot{\mathbf{v}} + \Delta\mathbf{v}^T R \Delta\mathbf{r} \tag{23}$$

To ensure that (23) is negative definite, it is equated to a negative definite function:

$$-\Delta\mathbf{v}^T S \Delta\mathbf{v} \tag{24}$$

where S is a positive definite matrix. Thus we have:

$$\dot{L} = \Delta\mathbf{v}^T m_F \Delta\dot{\mathbf{v}} + \Delta\mathbf{v}^T R \Delta\mathbf{r} = -\Delta\mathbf{v}^T S \Delta\mathbf{v} \tag{25}$$

Simplifying (25) and solving for the follower acceleration, we have:

$$m_F \dot{\mathbf{v}}_F = m_F \dot{\mathbf{v}}_L + R\Delta\mathbf{r} + S\Delta\mathbf{v} \tag{26}$$

Neglecting the disturbances from the slung load system, the nonlinear translational equations of the follower quadrotor may be written in the inertial frame as:

$$m_F \dot{\mathbf{v}}_F = -\begin{bmatrix} \tilde{k}_x & 0 & 0 \\ 0 & \tilde{k}_y & 0 \\ 0 & 0 & \tilde{k}_z \end{bmatrix} \begin{Bmatrix} v_{x_F}^2 \\ v_{y_F}^2 \\ v_{z_F}^2 \end{Bmatrix} + \begin{Bmatrix} F_{xC} \\ F_{yC} \\ F_{zC} \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ m_F g \end{Bmatrix} \tag{27}$$

Substituting (27) into (26) and solving for the thrust vector direction, we obtain:

$$\begin{Bmatrix} F_{xC} \\ F_{yC} \\ F_{zC} \end{Bmatrix} = \begin{bmatrix} \tilde{k}_x & 0 & 0 \\ 0 & \tilde{k}_y & 0 \\ 0 & 0 & \tilde{k}_z \end{bmatrix} \begin{Bmatrix} v_{x_F}^2 \\ v_{y_F}^2 \\ v_{z_F}^2 \end{Bmatrix} - \begin{Bmatrix} 0 \\ 0 \\ m_F g \end{Bmatrix} + m_F \dot{\mathbf{v}}_L + R\Delta\mathbf{r} + S\Delta\mathbf{v} \tag{28}$$

The control law obtained in (28) above shall realise the follower quadrotor prescribed formation geometry. The orientation of the thrust vector will be realised by the inner loop controller.

3.3. Mechanization of the attitude control

The attitude of the quadrotor is controlled by the torque and thrust of the propellers. In this manuscript, attitude control is realized using the *to-go* quaternion. Defining the current attitude with q and the desired attitude with d, the *to-go* quaternion, t, is defined as $d = q \otimes t$, then, $t = d \otimes t^{-1}$ [31].

In the body fixed coordinates of the quadrotor, the thrust vector direction is perpendicular to the quadrotor plane, (body fixed xy-plane) where superscript B indicates the body fixed frame coordinates:

$$\gamma^B = -\mathbf{k} \tag{29}$$

The navigation frame such as the North-East-Down frame may be assumed as the inertial frame. Thus, the desired thrust vector direction associated unit vector, dictated by the outer loop control algorithm is given as:

$$\lambda^N = \frac{F_{xC}^N \mathbf{I} + F_{yC}^N \mathbf{J} + F_{zC}^N \mathbf{K}}{F_{total}} \tag{30}$$

where superscript N indicates the navigation frame and,

$$F_{total} = \sqrt{(F_{xC}^N)^2 + (F_{yC}^N)^2 + (F_{zC}^N)^2} \tag{31}$$

The desired thrust vector direction can then be transformed to the body fixed frame using quaternion multiplication or the transformation matrix obtained from attitude propagation:

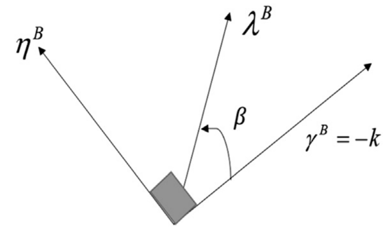

Fig. 5. Current Thrust Vector, γ^B, Desired Thrust Vector, λ^B, Rotation Axis, η^B, Rotation Angle, β.

$$\lambda^B = q^{-1} \otimes \lambda^N \otimes q = C_N^B \lambda^N \tag{32}$$

To rotate the current thrust vector direction to the desired thrust vector direction, the associated rotation axis and angle are needed. Fig. 5. The rotation axis η^B around which the quadrotor is rotated as well as the rotation angle β may be found thus:

$$\gamma^B \times \lambda^B = -\mathbf{k} \times \lambda^B = \eta^B \sin(\beta) \tag{33}$$

and

$$\gamma^B \cdot \lambda^B = -\lambda_z^B = \cos(\beta) \tag{34}$$

Then the *to-go* quaternion may now be obtained as follows:

$$s_4 = \cos(\beta/2) = \sqrt{\frac{1 - \lambda_z^B}{2}} \tag{35}$$

$$\mathbf{s} = \eta^B \sin(\beta/2) = \frac{-\mathbf{k} \times \lambda^B}{2s_4} = \frac{\lambda_y^B \mathbf{i} - \lambda_x^B \mathbf{j}}{2s_4} \tag{36}$$

We should also include the desired yaw motion of the vehicle. This could be done using another quaternion:

$$\begin{aligned} \mathbf{y} &= \mathbf{k}\sin(\Delta\psi/2) \\ y_4 &= \cos(\Delta\psi/2) \end{aligned} \tag{37}$$

where $\Delta\psi = \psi_{commanded} - \psi$, is the additional yaw rotation required from current attitude.

After including the additional yaw rotation, the total *to-go* quaternion may then be obtained by concatenating the quaternion associated with thrust vector direction and the quaternion associated with the additional yaw rotation:

$$t = s \otimes y \tag{38}$$

3.4. Inner loop quaternion based attitude controller design

The attitude of the quadrotor is controlled by pitching and rolling the quadrotor platform to align the total force of the quadrotor to the thrust vector direction from the LQT velocity/formation guidance controller. The quadrotor is rotated using the *to-go* quaternion obtained in Eq. (38). Again, neglecting the disturbances, the rotational equations of motion of the quadrotor can be written as [1]:

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} (I_x - I_z)qr/I_x \\ (I_z - I_x)pr/I_y \\ (I_x - I_y)pq/I_z \end{bmatrix} + \begin{bmatrix} d/I_x & 0 & 0 \\ 0 & d/I_y & 0 \\ 0 & 0 & 1/I_z \end{bmatrix} \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix} \tag{39}$$

A nonlinear quaternion based attitude controller using the Lyapunov stability theorem is then designed.

Let $S = \begin{bmatrix} (I_x - I_z)qr/I_x \\ (I_z - I_x)pr/I_y \\ (I_x - I_y)pq/I_z \end{bmatrix}$, $G = \begin{bmatrix} d/I_x & 0 & 0 \\ 0 & d/I_y & 0 \\ 0 & 0 & 1/I_z \end{bmatrix}$ and $T = \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix}$

Table 1
Properties of the quadrotors, load, and bars.

Bodies	Mass, Kg	Length, m	Moment of Inertia, Kg m^2
Leader/Follower Quadrotor	0.65	0.23 (Arm)	$\begin{bmatrix} 7.5\times10^{-3} & 0 & 0 \\ 0 & 7.5\times10^{-3} & 0 \\ 0 & 0 & 1.3\times10^{-2} \end{bmatrix}$
Cable	0.01	1	$\begin{bmatrix} 8.3\times10^{-4} & 0 & 0 \\ 0 & 8.3\times10^{-4} & 0 \\ 0 & 0 & 1.0\times10^{-7} \end{bmatrix}$
Load	0.2	0.5 (Radius)	$\begin{bmatrix} 2.0\times10^{-2} & 0 & 0 \\ 0 & 2.0\times10^{-2} & 0 \\ 0 & 0 & 2.0\times10^{-2} \end{bmatrix}$

Table 2
Designed controller settling times for the rigid bar case.

Formation guidance controller, s (**Outer Loop**) Follower Quadrotor	LQT Velocity Controller, s (**Outer Loop**) Leader Quadrotor	Attitude Controller, s (**Inner Loop**)
2	1.47	0.38
2	0.10	0.38
1	1.47	0.38

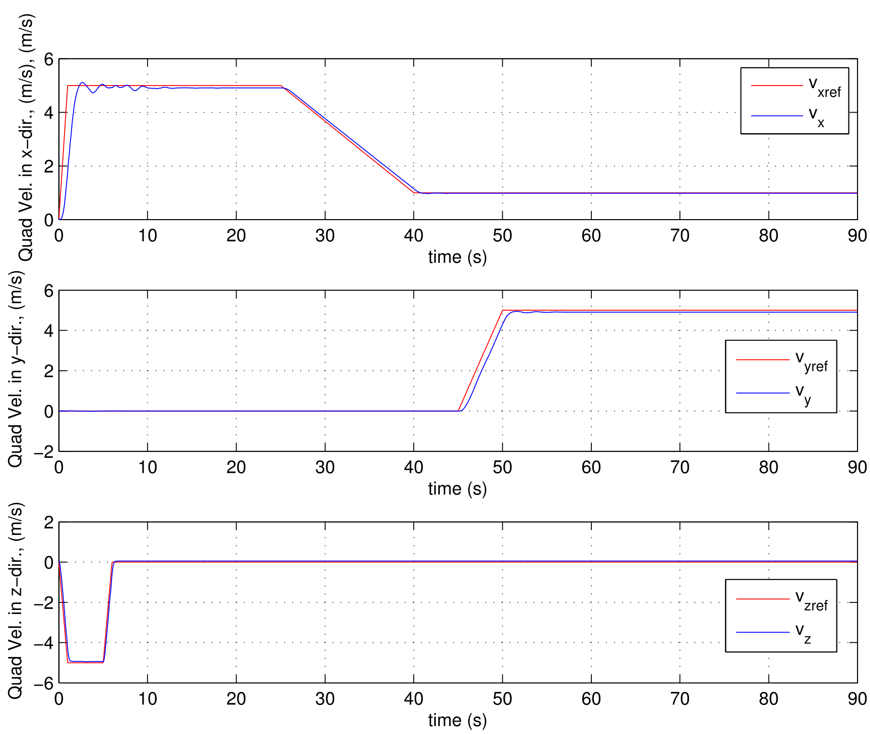

Fig. 6. Leader quadrotor velocity vector components along the x, y and z directions respectively. (For interpretation of the colours in the figure(s), the reader is referred to the web version of this article.)

Also, consider the following Lyapunov function [32]:

$$D = \frac{1}{2}\boldsymbol{\omega}^T M^{-1} J\boldsymbol{\omega} + 2(1 - t_4) \tag{40}$$

It is assumed that a positive definite M^{-1} exists and $M^{-1}J$ is also a positive definite symmetric matrix. Then the derivative of the Lyapunov function becomes:

$$\dot{D} = \boldsymbol{\omega}^T M^{-1} J\dot{\boldsymbol{\omega}} - \boldsymbol{\omega}^T \mathbf{t} \tag{41}$$

since $\dot{t}_4 = \frac{1}{2}\omega^T \mathbf{t}$. Choosing the decay rate as $\dot{D} = -\boldsymbol{\omega}^T M^{-1} N\boldsymbol{\omega}$, where $M^{-1}N$ is a positive definite matrix, ensures that the Lyapunov function, D keeps decreasing. Substituting Eq. (39) into Eq. (41), equating it to the decay rate and making the necessary simplifications, the following control law for the attitude control of the quadrotor may be obtained:

$$T = G^{-1}[J^{-1}(M\mathbf{t} - N\boldsymbol{\omega}) - S] \tag{42}$$

Using the control law obtained in Eq. (42) and making $F_{total} = \sqrt{(F^N_{xC})^2 + (F^N_{yC})^2 + (F^N_{zC})^2}$ where F_{total} is the magnitude of the thrust vector obtained from the LQT velocity/formation guidance controller, the rotor speeds to control the translational and rotational motion of the quadrotor can be obtained. Assuming that the torque and thrust of the propellers are proportional to the square of the rotor speeds, the following equations are written:

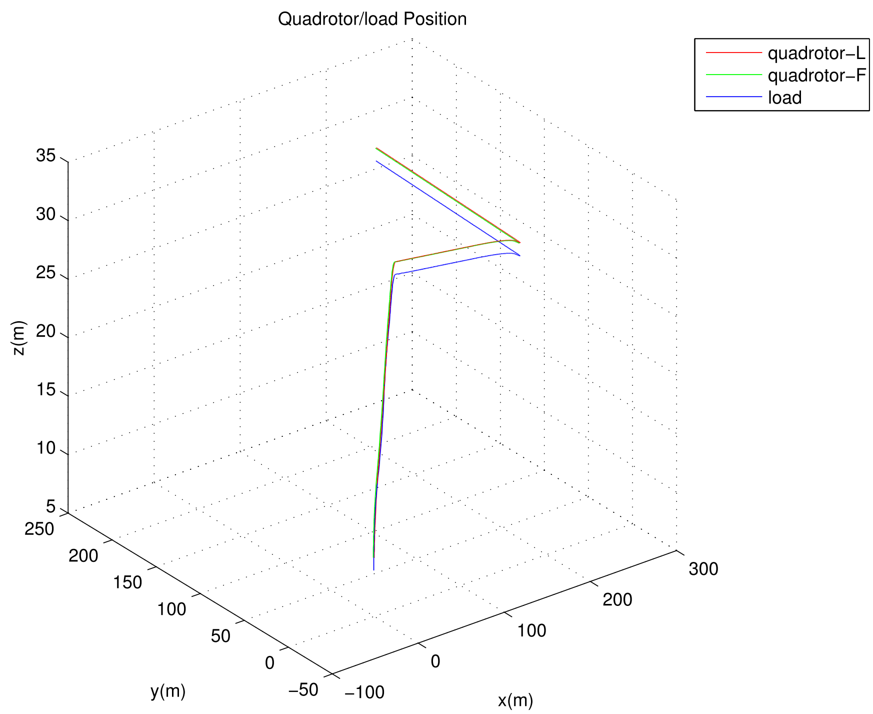

Fig. 7. Leader and follower quadrotor and slung load trajectories.

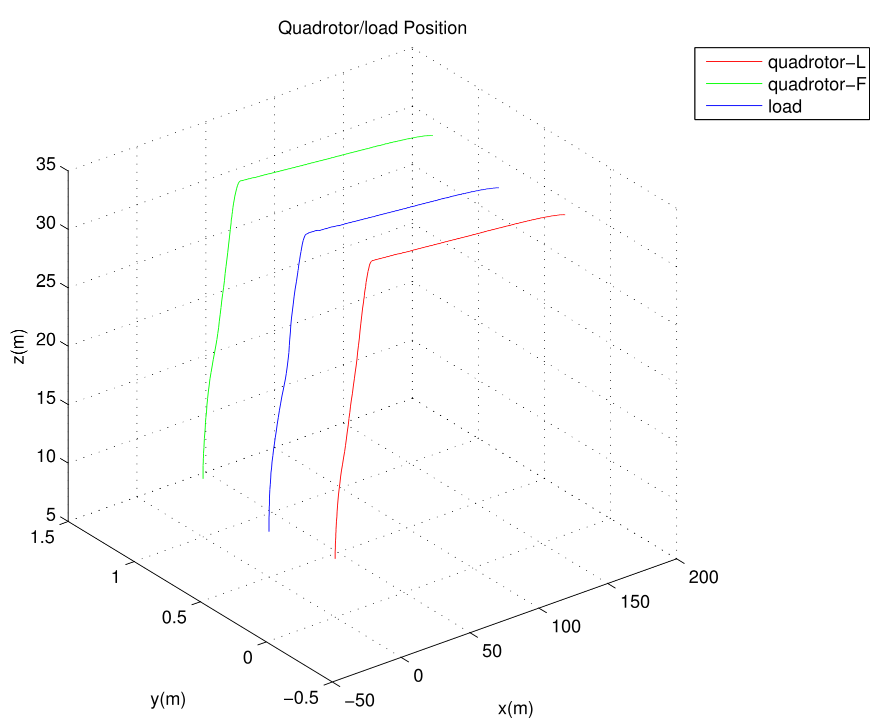

Fig. 8. Leader and follower quadrotor and slung load trajectories (First 40 seconds of the simulation).

$$\begin{Bmatrix} F_{total} \\ T_x \\ T_y \\ T_z \end{Bmatrix} = \begin{bmatrix} k_n & k_n & k_n & k_n \\ 0 & -k_n & 0 & k_n \\ k_n & 0 & -k_n & 0 \\ k_n k_m & -k_n k_m & k_n k_m & -k_n k_m \end{bmatrix} \begin{Bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{Bmatrix} \tag{43}$$

Ω_i^2 are the rotor angular velocities of the ith propeller of the quadrotors. In the simulink simulation model, these rotor angular velocities are passed through a first order low pass filter to simulate the dynamics of a DC motor [33].

$$\frac{20}{s+20} \tag{44}$$

The filtered rotor speeds is then used to obtain the rotor force and torque to control the quadrotor.

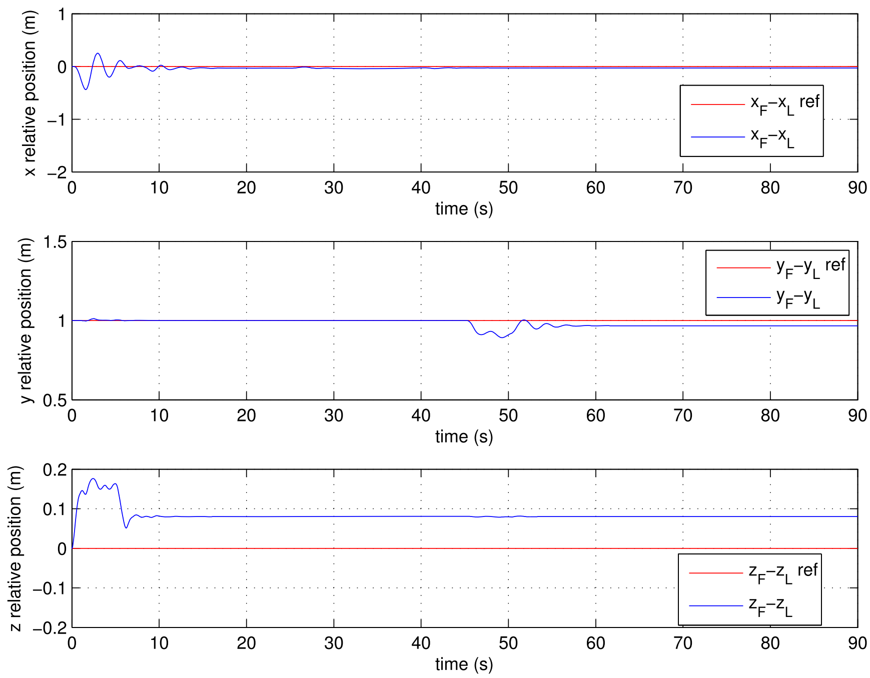

Fig. 9. Time history of the follower quadrotor position relative to the leader.

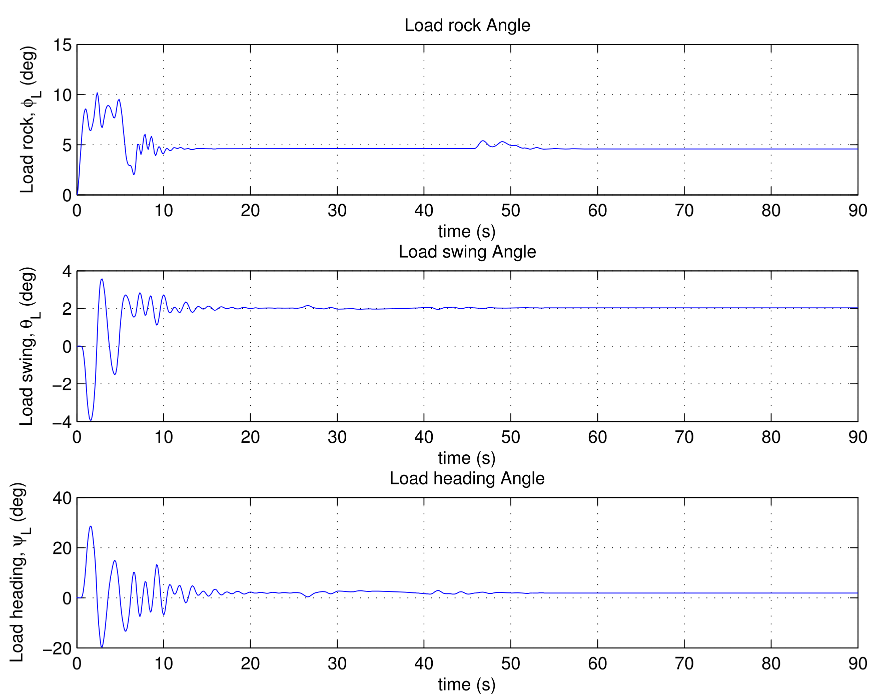

Fig. 10. Swing, rock and heading angles of the slung load during simulation.

4. Results and discussion

Using all the equations presented, a nonlinear simulation code is developed for the two quadrotor slung load system. The algorithms proposed are then tested against the developed model. To test the performance of our control system, the leader velocity commands are applied to the leader quadrotor, and the relative distance and position error commands are generated and sent to the formation guidance controller to enable the follower quadrotor maintain the desired relative distance. It is desired that the initial formation geometry be maintained in spite of the leader quadrotor's motion. The relative desired distance to be maintained is: $(x_F - x_L, y_F - y_L, z_F - z_L) = (0, 1, 0)$ m. The properties of the quadrotors and slung load system are listed in Table 1. Desired controller settling times for the formation controller, leader quadrotor controller and inner loop attitude controller are listed in Table 2. Accordingly, the outer loop settling times are higher than the attitude controllers. The leader quadrotor controller on the other hand is slightly slower than the formation controller.

Fig. 6 presents the leader quadrotor response to the velocity commands. From the figure, it may be observed that the leader

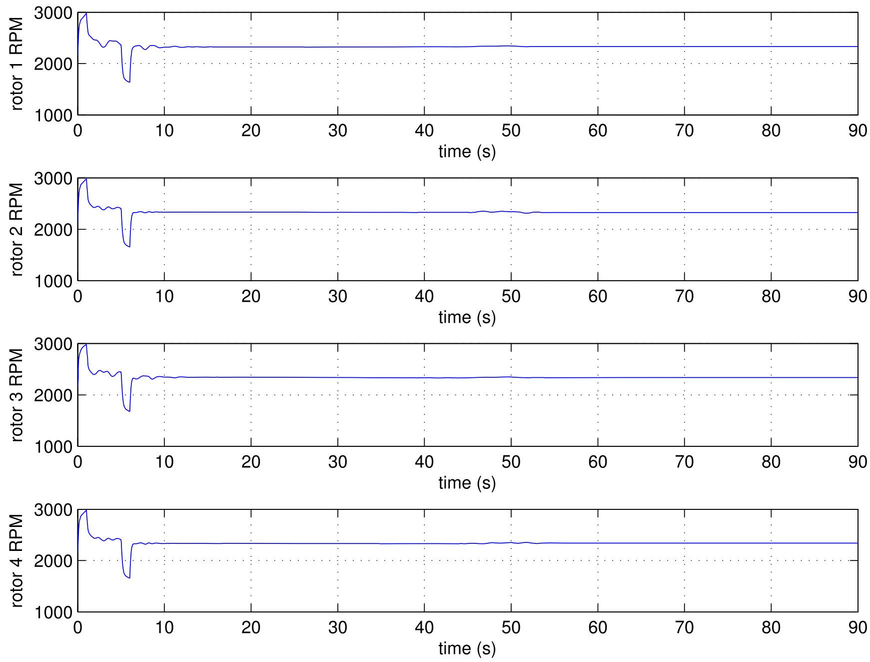

Fig. 11. Realized propeller speeds of the leader quadrotor.

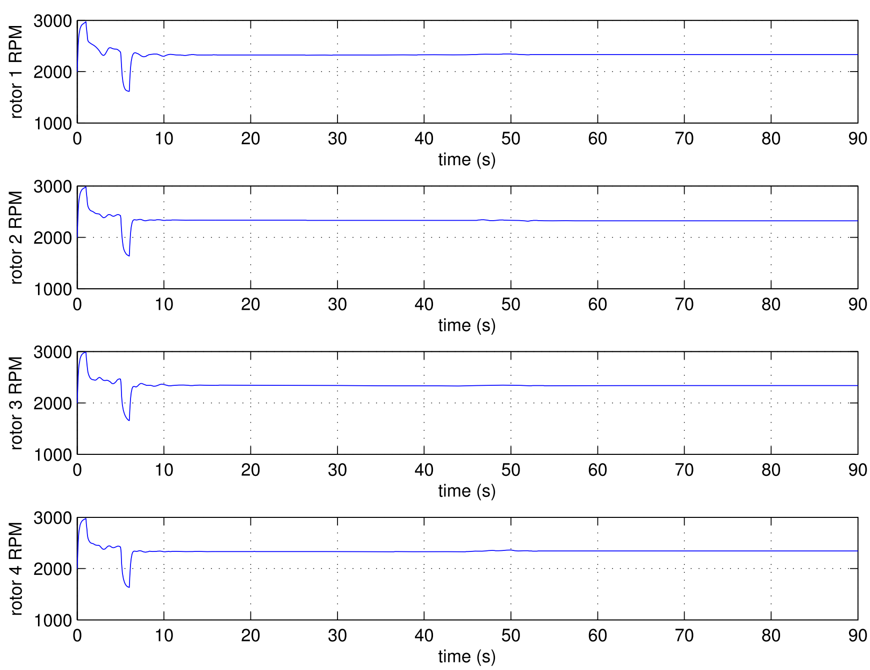

Fig. 12. Realized propeller speeds of the follower quadrotor.

quadrotor closely follows the desired velocity commands in spite of the disturbance effect of the slung load. Fig. 7 and 8 show the position flight trajectories. The follower quadrotor is able to keep up with the leader quadrotor for the duration of the flight. Fig. 9 shows the time history of the position of the follower quadrotor with respect to the leader quadrotor. There is some tracking error reaching up to 0.18 m at the beginning and when the velocity change is requested from the leader quadrotor. However, these disappear rapidly and the quadrotors settle to the desired relative configuration.

Fig. 10 shows the load rock, swing and heading angles. The two quadrotors are able to carry the load in a smooth fashion. We see a deflection in the load rock angle from approximately 5 degrees to 10 degrees in the first 10 seconds and then settles to about 5 degrees for the remaining duration of the flight. The swing angle of the load oscillates between -4 degrees and 4 degrees in the first 10 seconds of simulation and then settles at 2 degrees. The heading angle also oscillates between -20 degrees and 30 degrees in the first 10 seconds. All these angles are quite small and may be considered acceptable for the package delivery operation. Fig. 11 and 12 show the realized propeller speeds of the leader and follower quadrotors during the flight. These speeds may be easily achieved and they do not reach the saturation limit of 5000 rpm envisaged.

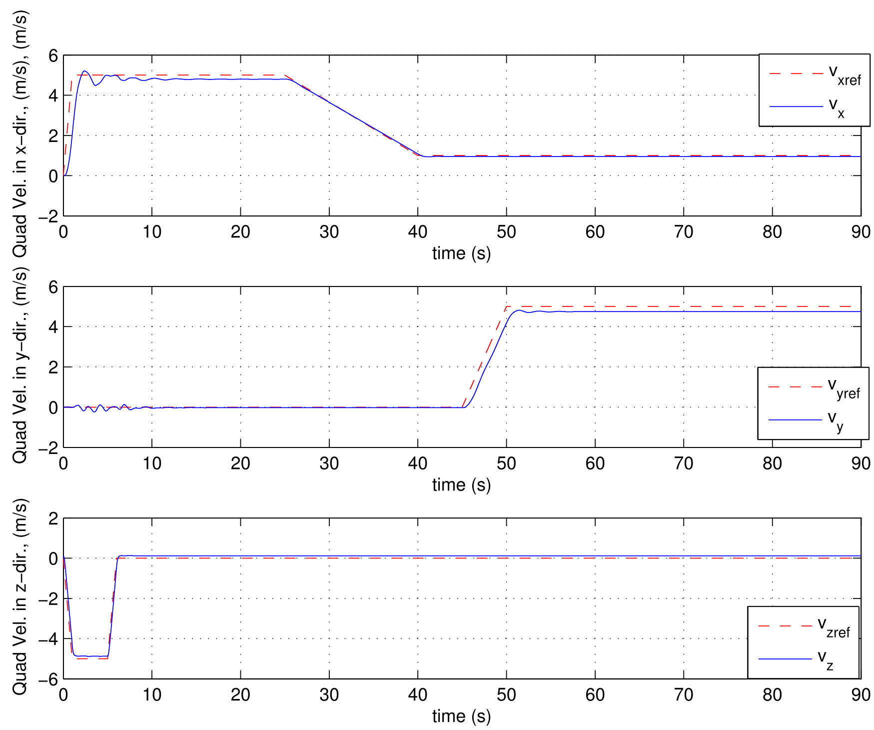

Fig. 13. Leader quadrotor velocity vector components along the x, y and z directions respectively (Flexible bar case).

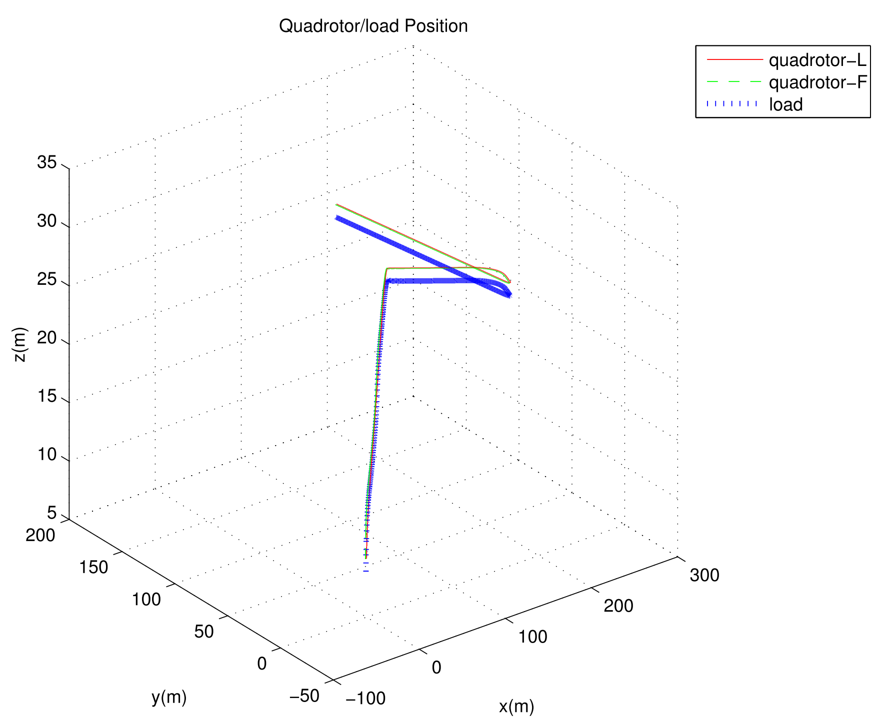

Fig. 14. Leader and follower quadrotor and slung load trajectories (Flexible bar case).

Table 3
Designed controller settling times for the flexible bar case.

Formation Guidance Controller, s (**Outer Loop**) Follower Quadrotor	LQT Velocity Controller, s (**Outer Loop**) Leader Quadrotor	Attitude Controller, s (**Inner Loop**)
1.38	0.69	0.38
0.69	0.04	0.38
0.69	0.85	0.38

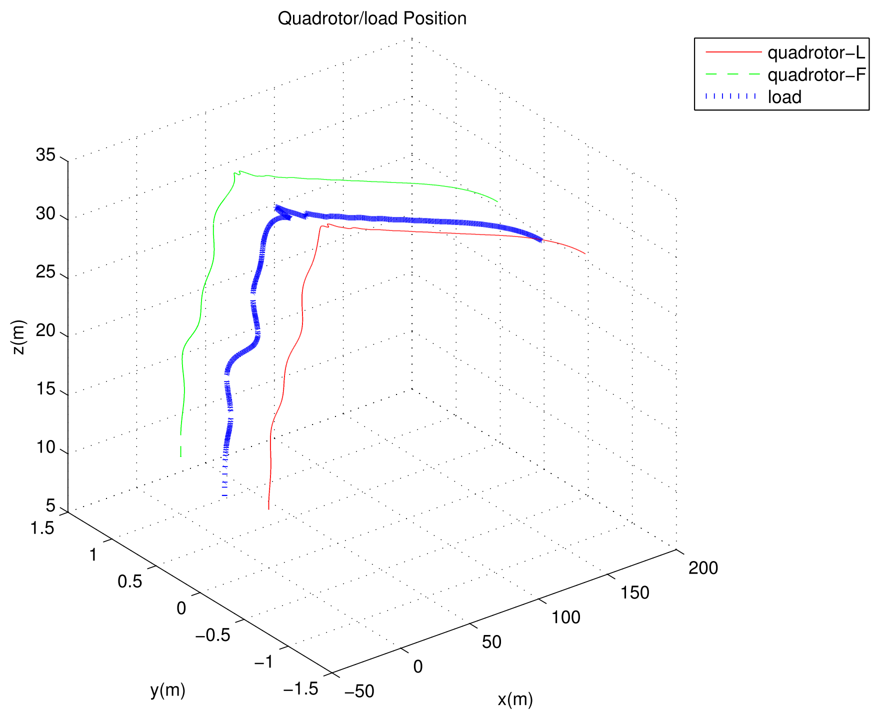

Fig. 15. Leader and follower quadrotor and slung load trajectories (First 40 seconds of the simulation, Flexible bar case).

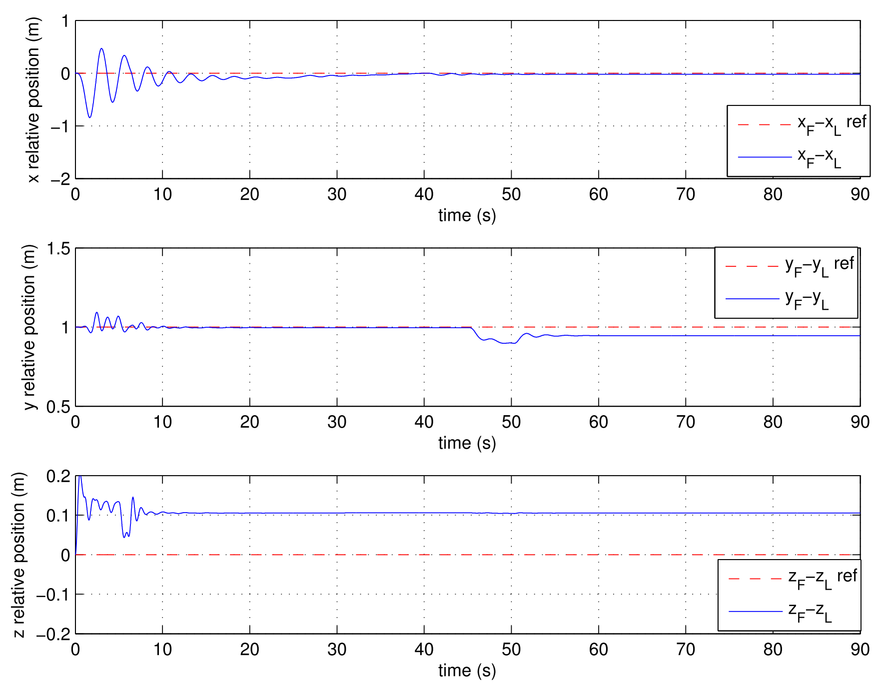

Fig. 16. Time history of the follower quadrotor position relative to the leader (Flexible bar case).

The above cases are repeated this time using the simulation model obtained from ADAMS software code embedded to the simulink simulation model, as described in the previous section. The same hierarchical control algorithms are used as before. The controllers are tuned to the settling times listed in Table 3. These settling times are smaller in the outer loops than the previous controller. The inner loop settling times are same as before. The simulation results are presented in Figs. 13–19.

Fig. 13 gives the commanded and realized velocities of the leader quadrotor. Its performance may be compared to the previous rigid bar simulation case. With the flexible bars, the leader displays slightly higher amplitude oscillations than before. Flight trajectories of the leader, follower, and the load are presented in Fig. 14 and 15. Comparing these figures with the rigid bar case (Fig. 7 and Fig. 8), it may be observed that the flexible bar simulation results are more oscillatory. Similarly, the follower quadrotor tracks the leader with higher oscillatory errors than the rigid bar case (Fig. 6) reaching up to 1 m error in the longitudinal axis. However, steady state error of about 0.1 to 0.2 m is observable along the lateral and vertical axes as it was the case with the

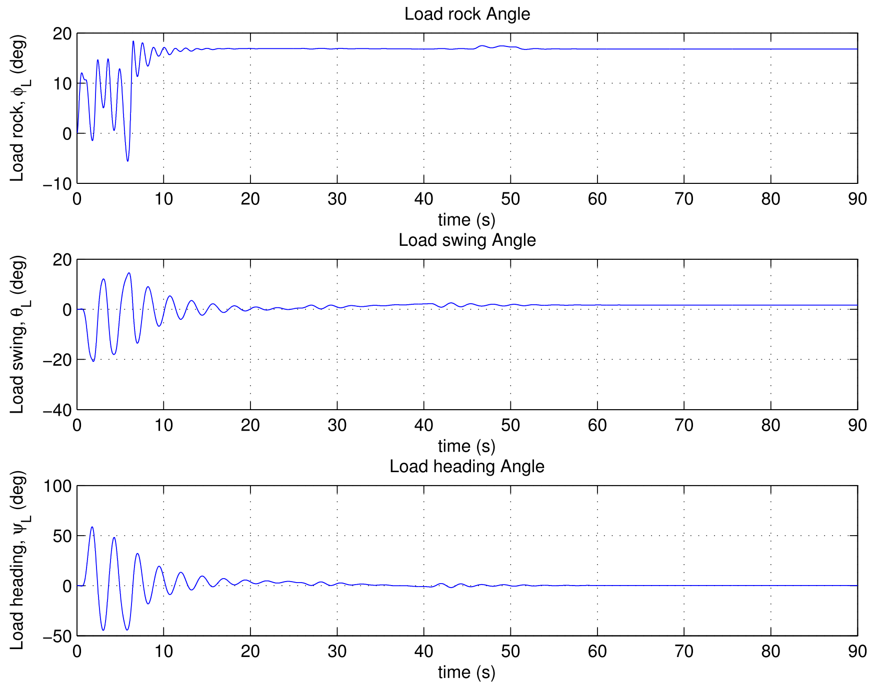

Fig. 17. Swing, rock and heading angles of the slung Load during simulation (Flexible bar case).

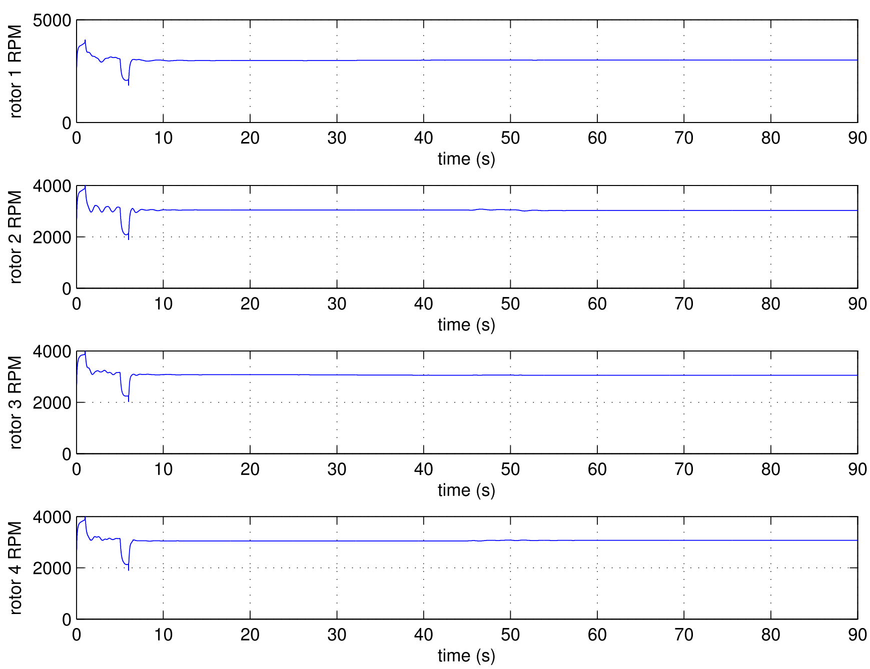

Fig. 18. Realized propeller speeds of the leader quadrotor (Flexible bar case).

previous simulation carried out assuming rigid bars. Higher rock, swing and heading angles may also be observed from Fig. 17. Again the slung load oscillations disappear rapidly as the steady flight is achieved. Steady state rock angle is also higher (about 18°) as compared to the previous case (about 5°). The propeller speeds reached, shown in Fig. 18 and 19 presents higher transients, reaching up to 4000 rpm, are still reasonable and may be easily realized by the quadrotors. From the simulations with the flexible bar, it may be concluded that this unmodelled dynamics in the design of the controller does not cause severe performance degradation or instability.

5. Conclusion

An autonomous control system for the formation flight of two quadrotors carrying a slung load is designed and implemented. The control algorithm features a two loop architecture in which a thrust vector is generated in the outer loop: Lyapunov function based formation guidance for the follower and LQT based velocity controller for the leader. The thrust vector direction is then used to generate a reference *to-go* quaternion for the inner loop nonlinear attitude controller. Both controllers are based on rigid body equations of motion of the quadrotors. Thus, the slung dynamics is

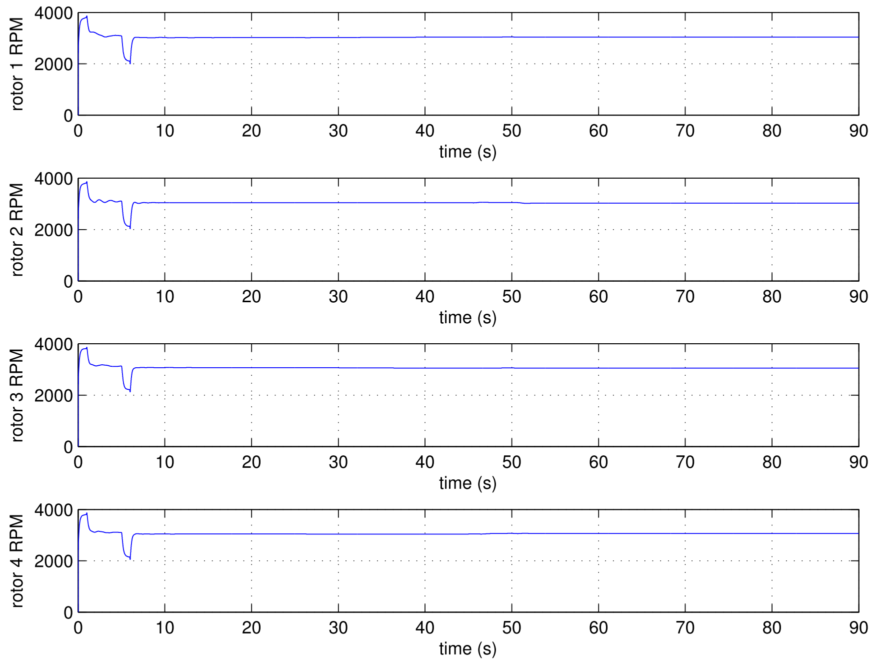

Fig. 19. Realized propeller speeds of the follower quadrotor (Flexible bar case).

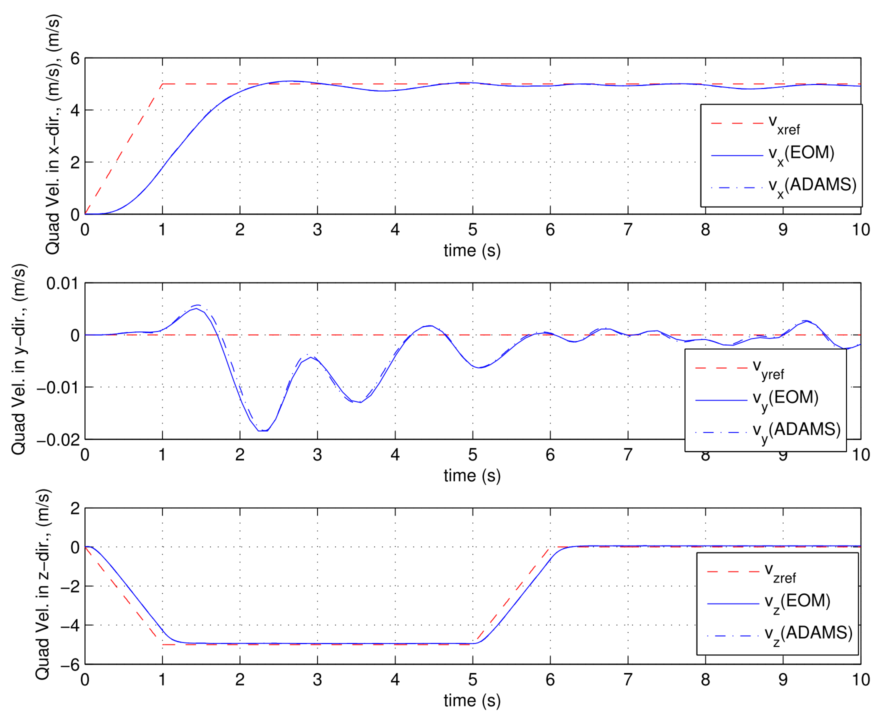

Fig. 20. Comparison between leader quadrotor velocities.

neglected in the controller designs. The simulation results show that the quadrotors are able to fly in formation under the disturbance of the slung load without becoming unstable or causing saturation in the propeller speeds. The developed control system produces satisfactory results for both the rigid and flexible bar cases.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Comparison of the simulation codes developed in simulink using rigid bar slung load model obtained using the equations of motion derived in section 2 and a rigid bar slung load model using the ADAMS software is presented in this appendix to give some level of verification to the simulation codes developed. The same feedback control law is used in both simulation models. The results are presented in Figs. 20–22. The leader quadrotor is commanded to follow a prescribed velocity vector. The results given in Fig. 20 shows that the simulated leaders follow the command closely in both simulations. Their responses are also quite close to each other.

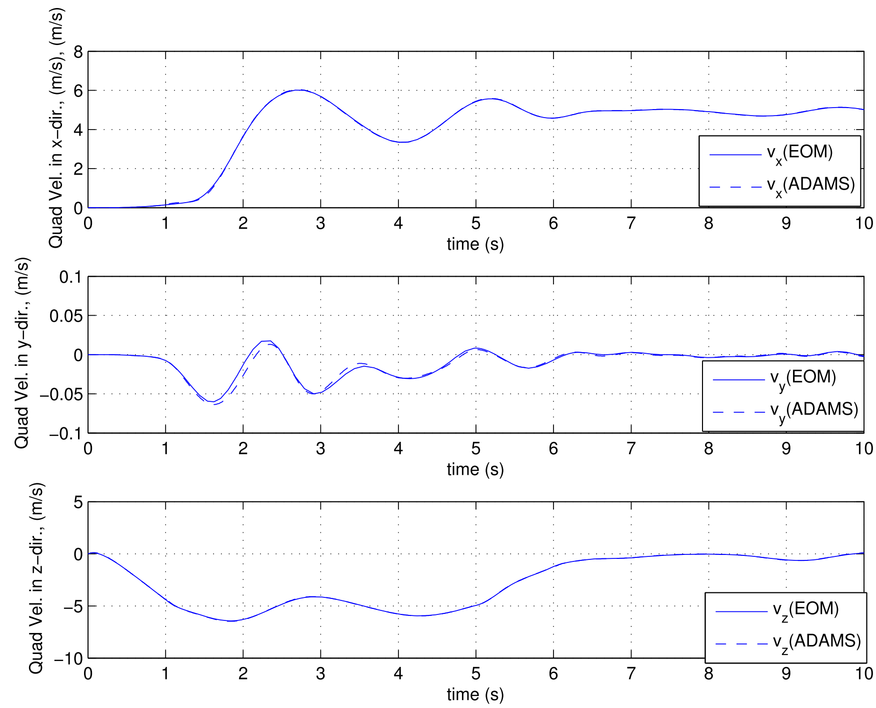

Fig. 21. Comparison between follower quadrotor velocities.

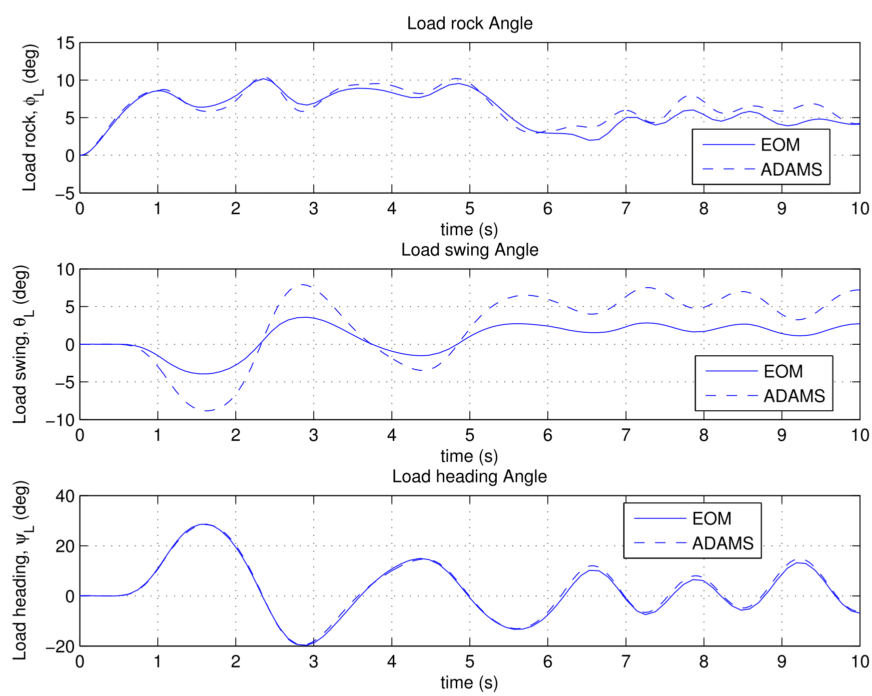

Fig. 22. Comparisons between load angle deflections.

Follower velocities are presented in Fig. 21. It may be observed that the follower velocities obtained from both codes are also close to each other. The load angle deflections obtained from these two simulations are also presented in Fig. 22. This figure also demonstrates that the simulation codes are giving similar results. It may be concluded that the ADAMS code embedded simulation is giving similar results to that of the simulation based on the equations presented in section 2.

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